

Conjecture Settling II

Automated Logical Deciders and Three-Way Lemma Exchange

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Developed and formalized in collaboration with ChatGPT

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Research objective

Build a certificate-producing machine that can settle a conjecture relative to named axioms by **proving it, refuting it, or proving its independence** - while allowing all three lines of attack to exchange verified lemmas safely.

Abstract

The settlement machine of *Conjecture Settling 001* searches simultaneously for a proof of a conjecture c and a proof of its negation, with verified lemmas shared between the two searches. This sequel adds the missing third outcome: certified independence of c from the object theory T . The resulting Automated Logical Decider (ALD) is not one universal decision algorithm. It is a certificate-producing, generally partial architecture containing object-level proof agents, model construction agents, a metatheoretic proof agent, a typed lemma federation, and a small traffic coordinator. The central proposal is three-way lemma exchange: a theorem, reduction, normal form, model fragment, or proof-theoretic bridge discovered on one route may accelerate the other routes, provided every transfer is checked in its destination theory. We define the architecture, prove a soundness theorem for typed sharing, identify direct syntactic, model-pair, consistency-pair, forcing, and interpretability routes to independence, generalize cross-branch reuse to a reuse matrix and a tri-use rate, connect scheduling to the creativity objective $Y = D N$, give a controlled measurement protocol, and state an implementation roadmap for Predator.

The principal warning is equally important: an ALD may report independence only from a formal certificate. Failure of the proof and refutation branches, no matter how prolonged, is never such a certificate.

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1. From a two-sided settlement machine to an ALD

An ordinary automated theorem prover is target-directed: it searches for a certificate of $T \vdash c$. The settlement machine introduced in *Conjecture Settling 001* duplicates the search, pursuing c and $\neg c$ while permitting closed, verified lemmas to cross the branch boundary. This changes failure into a more informative experiment. If the negative branch succeeds, the conjecture is refuted; if the positive branch succeeds, it is proved. The nonhalting branch can still contribute lemmas to the branch that eventually settles the target.

Two branches are nevertheless incomplete as an account of mathematical settlement. A conjecture may be neither provable nor refutable from T . Cohen did not give up on the Continuum Hypothesis when neither direction yielded a ZFC proof; the logical status itself became the theorem. The next machine must therefore be able to change levels. It must reason not only *inside* T , but *about* T in a named metatheory M .

ATP: $T \vdash c$? Settlement machine: $T \vdash c$ or $T \vdash \neg c$?

ALD: What is the certified logical status of c relative to T ?

Definition 1.1 (Automated Logical Decider)

An **Automated Logical Decider** is an orchestrated collection of formal search procedures that, given an object theory T , a sentence c , a metatheory M , and a resource budget B , attempts to return a checkable certificate for PROVED, REFUTED, INDEPENDENT, or INCONSISTENT. If no such certificate is found, it returns an explicitly operational result such as TIMEOUT, OVERFLOW, FAULT, or INTERRUPTED. Operational failure carries no theorem-level meaning.

The word *decider* ordinarily denotes a total halting procedure in computability theory. For sufficiently strong theories no such universal total ALD can exist. The general machine proposed here is therefore a **certificate-producing partial ALD**. The shorter name describes its task; the qualifier describes its mathematically possible implementation.

1.1 Contribution of this sequel

- A minimal three-Predator ALD and a full five-Predator-plus-coordinator ALD.
- A typed lemma federation replacing the unsafe idea of one unlabelled global pool.
- Direct, model-theoretic, consistency, forcing, and interpretation routes to independence.
- A proof that destination-verified lemma transport preserves soundness.
- A reuse matrix, a three-way reuse rate, fan-out, and branch-specific settlement dividends.
- A correction to any unconditional claim that sharing cannot delay heuristic search.
- A falsifiable protocol for testing whether the three-way architecture and $Y = D \ N$ improve solves.

1.2 Related work and the honest novelty claim

Parallel and cooperative theorem proving, blackboard lemma exchange, strategy portfolios, model finding, and multi-agent proof construction all have precedents. Parallel Modoc, for example, ran multiple processes that shared lemmas and autarkies through a blackboard [12]. Modern agentic proof systems can generate auxiliary lemmas under formal feedback [13], and proof/model-checking systems can return either certificates or counterexamples in specialized logics [14]. Therefore this paper does not claim priority for 'several provers talking' or for proof-versus-model search by itself.

The potentially distinctive contribution is the combination: an explicit three-status objective {proof, refutation, relative independence}; destination-verified sharing across object theory, opposite model contexts, and metatheory; quantitative tri-use and settlement-dividend measurements; and a creativity-guided population evaluated under anti-leakage rules. Even that novelty claim should remain provisional until a dedicated literature review and working comparison are complete. What will make the work consequential is not the name ALD but a verified result that the architecture reaches and explains statuses that matched baselines do not.

2. Formal status and certificates

Fix an effectively presented object theory T and a sentence c in its language. Let M be a theory capable of formalizing enough syntax, proof theory, and semantics for the proposed independence argument. For set-theoretic independence M may itself be a stronger set theory; for finite first-order examples, M may formalize model checking and first-order soundness.

Definition 2.1 (Decided and independent)

The sentence c is **decided by T** when $T \vdash c$ or $T \vdash \neg c$. It is **independent of T** when $T \nvdash c$ and $T \nvdash \neg c$. Independence is always relative to the named theory. An ALD's independence report is also relative to the metatheory in which the certificate has been verified.

MATHEMATICAL STATUS	MINIMUM CERTIFICATE	INTERPRETATION
PROVED	A verifier-accepted T -proof of c .	$T \vdash c$.
REFUTED	A verifier-accepted T -proof of $\neg c$.	$T \vdash \neg c$.
INDEPENDENT	An M -certificate of both unprovability directions, or a certified construction implying them.	$T \nvdash c$ and $T \nvdash \neg c$, relative to M .
INCONSISTENT	A verifier-accepted T -proof of contradiction.	The object theory proves false; proof and refutation cease to be exclusive.

2.1 Mathematical outcomes versus operational outcomes

A benchmark must not merge mathematics with process termination. PROVED, REFUTED, INDEPENDENT, and INCONSISTENT are mathematical outcomes because each carries a certificate. TIMEOUT means the fixed budget expired. OVERFLOW means an explicit resource limit was exceeded. FAULT means the software or verification pipeline failed. INTERRUPTED means continuity was externally broken. None of the latter four is evidence for any logical status.

2.2 Mutual exclusivity and inconsistency

If T is consistent, PROVED, REFUTED, and INDEPENDENT are mutually exclusive. If accepted certificates support both c and $\neg c$, the ALD should not choose whichever arrived first; it should assemble a contradiction certificate and report INCONSISTENT. Likewise, a proof of c and a sound model of $T + \neg c$ cannot coexist. Such a collision is a diagnostic event: at least one certificate, transport, formalized soundness assumption, or theory identifier is wrong.

Non-negotiable rule

Search exhaustion is evidence only about the search procedure and its budget. Independence requires a separate positive certificate. A billion failed proof attempts do not become a proof of unprovability.

3. The Predators and the traffic coordinator

3.1 Minimal three-Predator architecture

The user's core design requires only three mathematical agents. Predator P+ searches inside T for c . Predator P- searches inside T for $\neg c$. Predator PI searches in M for a certificate of independence. The first two can share every closed theorem of T . PI may consume those proofs as coded facts about T , and may return bridge lemmas, reductions, model obligations, or consistency obligations. The important qualification is that the three agents do not inhabit one undifferentiated logical world.

3.2 Full five-Predator architecture

Independence naturally splits into two semantic halves and one assembly task. This gives five specialists. A sixth component directs traffic but is deliberately not a theorem prover.

PREDATOR	FORMAL CONTEXT	PRIMARY TARGET	SUCCESS CONTRIBUTION
P+	Object theory T	$T \vdash c$	Immediate PROVED certificate
P-	Object theory T	$T \vdash \neg c$	Immediate REFUTED certificate
PM+	Models of $T + c$	Construct and certify $A \models T + c$	Shows $T \not\models \neg c$
PM-	Models of $T + \neg c$	Construct and certify $B \models T + \neg c$	Shows $T \not\models c$
PI	Metatheory M	Assemble unprovability, consistency, model, forcing, or interpretation certificates	Final INDEPENDENT certificate

Five mathematical agents exchange proof-carrying packets through a typed federation:

P+ prove c	P- prove not- c	PM+ model $T+c$	PM- model $T+\text{not-}c$	PI prove independence
TRAFFIC COORDINATOR: schedule - verify - type - route - assemble - log				

3.3 What the coordinator is allowed to do

The coordinator is a message router and scheduler. It allocates budgets, sends goals and verified lemmas to agents, maintains theory identifiers and dependency graphs, detects complete certificate combinations, and records measurements. It may rank or prioritize search opportunities, but it does not assert mathematical truth. If its scheduling heuristic is wrong, the run may be slow; it should not become unsound.

3.4 What the coordinator must never do

- Treat a model candidate as a model certificate before all axioms and the target sign are checked.
- Insert a theorem of M into T merely because the formula looks similar.
- Share branch substitutions or live metavariable bindings; only closed results or typed proof states may cross.
- Infer independence from low search probability, elapsed time, or repeated failure.
- Hide a timeout, overflow, fault, or interruption under the word UNSETTLED.

4. The typed lemma federation

The original two-branch machine can share a ground theorem because both branches prove consequences of the same Γ in the same formal system. An ALD crosses theory levels. One global list of formulas is therefore unsafe. The replacement is a federation of stores whose entries carry their context, certificate, origin, and permitted transports.

Definition 4.1 (Proof-carrying lemma packet)

A lemma packet is a tuple $L = (\text{id}, \text{context}, \text{statement}, \text{certificate}, \text{origin}, \text{dependencies}, \text{cost}, \text{interfaces})$. The context names the exact theory or semantic environment. The certificate is checked before insertion. Interfaces list registered destination translations; a translation is accepted only when the destination verifier checks the translated certificate.

4.1 Principal contexts

CONTEXT	EXAMPLE CONTENT	DEFAULT IMPORT RULE
Obj(T)	Formal theorems $T \vdash \phi$	May weaken into $\text{Obj}(T+c)$ and $\text{Obj}(T+\neg c)$; may be coded in M as a verified proof fact.
Obj(T+c)	Consequences using the added assumption c	Cannot descend to $\text{Obj}(T)$ without discharging c ; cannot cross to $\text{Obj}(T+\neg c)$.
Obj(T+$\neg c$)	Consequences using the added assumption $\neg c$	Cannot descend to $\text{Obj}(T)$ without discharging $\neg c$; cannot cross to $\text{Obj}(T+c)$.
Meta(M)	Claims about proofs, consistency, models, interpretations, or forcing	Stays in M unless a separately verified reflection or interpretation rule applies.
Model(A)	A structure, valuation, satisfaction facts, finite tables	May support $\text{Meta}(M)$ soundness arguments; it is not automatically an object-theory theorem.

4.2 Safe transports

The most important safe transport is weakening: a proof from T remains a proof when an extra axiom is added, so an $\text{Obj}(T)$ theorem may be imported into $\text{Obj}(T+c)$ and $\text{Obj}(T+\neg c)$. The reverse direction is not valid. A proof of ϕ from $T+c$ can, when a formal deduction theorem is available, be transformed into a proof of $c \rightarrow \phi$ from T ; it does not become a proof of ϕ from T .

A concrete T -proof p of ϕ can be reflected upward as a metatheoretic claim $\text{Proof}_T(p, \ulcorner \phi \urcorner)$, because M can check the finite proof code. Under an appropriate definition, M may then derive $\text{Prov}_T(\ulcorner \phi \urcorner)$. The converse step from a metatheoretic theorem to an object theorem is not free: it would require a soundness or reflection principle whose strength must be explicit.

Theorem 4.2 (Soundness of destination-verified transport)

Assume each store accepts a packet only after its own verifier checks the packet's certificate in the packet's declared destination context. Then every accepted packet is a theorem or semantic fact of its declared context, regardless of how unreliable the producing agent, translator, scheduler, or language model may be.

Proof. Order accepted packets by insertion time. Initial packets are checked by the destination verifier. At a later insertion, the verifier checks the complete destination certificate using only the destination axioms and previously accepted destination packets. By induction those dependencies are valid in the destination context, so the new packet is valid there as well. The producer's internal reasoning is irrelevant. $\square$

4.3 Why substitutions do not cross

A live substitution belongs to a local proof state. The symbol x in one branch need not denote the same metavariable, have the same scope, or satisfy the same freshness constraints in another. Sharing unification environments can silently identify unrelated variables. The safe unit is the closed, fully scoped lemma together with its proof, or a deliberately typed subgoal interface whose transport is checked. This extends the warning in Remark 13.2 of *Conjecture Settling 001*.

5. What can be useful to all three lines of attack?

The most exciting possibility is not merely that $P+$ and $P-$ share ordinary theorems. A lemma can have three-way utility: it can simplify the proof of c , simplify the proof of $\neg c$, and simplify the metatheoretic or model-theoretic demonstration that neither proof exists. The same formula is not blindly copied everywhere; its certified consequences are transported through the interfaces of Section 4.

5.1 Five classes of high-value shared lemmas

- **Core theorems of T.** Algebraic, logical, set-theoretic, or structural results available to both object branches and formalizable as proof facts in M .
- **Equivalences and reductions.** A theorem $T \vdash (c \leftrightarrow d)$ moves all three status questions to a potentially easier sentence d .

- **Normalization and definitional lemmas.** Canonical forms reduce proof branching, model constraints, and the complexity of encoded proof predicates.
- **Bridge lemmas.** Deduction, soundness, completeness, conservativity, interpretation, forcing, and satisfaction lemmas connect object proofs with metatheoretic obligations.
- **Bounded negative certificates.** An exhaustive, verified check that no proof of length at most L exists can prune a bounded branch and inform PI, while never being confused with unbounded unprovability.

5.2 The equivalence bridge

Proposition 5.1 (A genuinely three-way lemma)

If $T \vdash (c \leftrightarrow d)$, then $T \vdash c$ iff $T \vdash d$; $T \vdash \neg c$ iff $T \vdash \neg d$; and c is independent of T iff d is independent of T . Models and consistency obligations transport accordingly.

Proof. Each object-level implication follows by composing a proof with one direction of the equivalence; the negated statements follow classically. Therefore provability and refutability agree for c and d , so their simultaneous absence agrees as well. Every model of T satisfies the equivalence, hence its truth value for c equals its truth value for d . $\square$

This proposition gives a precise realization of the user's idea. P+ may discover an equivalence while trying to prove c . P- can immediately redirect toward $\neg d$. PM+ and PM- can search for models with d true or false instead of c . PI can transfer an independence theorem for d back to c . One verified theorem changes the geometry of every branch.

5.3 Sharing can prune as well as accelerate

A certified model of $T + \neg c$ closes the unprovability half $T \nvdash c$. At that point P+ is no longer merely unlikely to succeed; relative to the accepted soundness certificate, success is impossible. The coordinator may stop P+ and redirect its budget. Symmetrically, a model of $T + c$ closes the unprovability of $\neg c$. A heuristic suspicion never permits this pruning; a certificate does.

6. Five routes to an independence certificate

6.1 Direct syntactic unprovability

$$M \vdash \neg \text{Prov}_T(\ulcorner c \urcorner) \wedge \neg \text{Prov}_T(\ulcorner \neg c \urcorner)$$

This is the most literal target for PI. It requires an arithmetization or other formal coding of T -proofs inside M . Direct unprovability proofs are often difficult, but they have the cleanest interface: the result already states the desired status.

6.2 A pair of models

$$M \vdash A \models T + c \quad \text{and} \quad M \vdash B \models T + \neg c$$

Proposition 6.1 (Model-pair independence)

Assume M verifies first-order soundness for T . If M verifies a model A of $T + c$ and a model B of $T + \neg c$, then M proves that c is independent of T .

Proof. If $T \vdash c$, soundness makes c true in every model of T , contradicting $B \models \neg c$. Thus $T \nvdash c$. If $T \vdash \neg c$, soundness contradicts $A \models c$. Thus $T \nvdash \neg c$. $\square$

6.3 A pair of consistency proofs

$$M \vdash \text{Con}(T + c) \wedge \text{Con}(T + \neg c)$$

If T proved c , adjoining $\neg c$ would make the extension inconsistent. If T proved $\neg c$, adjoining c would make the other extension inconsistent. Therefore consistency of both extensions yields both unprovability directions. The certification is relative to M , and the ability of M to prove those consistency statements is itself constrained by incompleteness.

6.4 Forcing and inner-model routes

For set theory, the semantic agents need not enumerate finite structures. They may construct an inner model or a forcing extension, prove that the relevant axioms are preserved, and prove opposite truth values for c . Goedel's constructible-universe argument and Cohen's forcing argument for the Continuum Hypothesis are the archetype [2, 3, 4]. Automating this route requires formal libraries for models of set theory, names, forcing relations, genericity, preservation, and the transfer of satisfaction claims. It is vastly more demanding than launching two ordinary ATPs.

6.5 Interpretability and conservativity

Sometimes the shortest route is neither a direct proof-code argument nor an explicit model. PI may interpret $T + c$ and $T + \neg c$ in theories already known consistent, prove a conservativity theorem, or reduce the target to a previously classified independent sentence. Such routes make equivalence and reduction lemmas especially valuable across all agents.

6.6 The metatheory cannot be hidden

An ALD must print the metatheory, translations, and relative assumptions alongside INDEPENDENT. The proper report is not simply ' c is undecidable.' It is, for example, ' $\mathcal{M}$ verifies that c is independent of T , via these two model certificates.' A stronger theory may decide the same sentence, and sufficiently strong consistent theories cannot in general prove every fact about their own proof-theoretic limitations [5].

7. A worked miniature ALD settlement

Let T_{grp} be the first-order axioms of groups and let c assert that every two elements commute. Neither c nor $\neg c$ follows from group theory alone. This example is small enough to display the five-agent logic without pretending that its formalization is comparable to a forcing proof.

AGENT	ATTEMPT	POSSIBLE RESULT
P+	Derive universal commutativity from group axioms.	Search does not settle; any derived group theorem enters $\text{Obj}(T_{\text{grp}})$.
P-	Derive existence of noncommuting elements from group axioms.	Search does not settle; it shares ordinary group lemmas with P+.
PM+	Build a group satisfying commutativity.	The two-element cyclic group C_2 supplies a finite model certificate.
PM-	Build a group violating commutativity.	The symmetric group S_3 supplies a finite nonabelian model certificate.
PI	Verify both tables satisfy the group axioms and opposite signs of c ; apply soundness.	Certifies that commutativity is independent of the group axioms.

The example clarifies what the independence Predator proves. It does not stare at the two unsuccessful proof searches and declare a draw. It produces two positive mathematical objects and a positive metatheorem: C_2 is a model of $T_{\text{grp}} + c$, S_3 is a model of $T_{\text{grp}} + \neg c$, and soundness converts the pair into two unprovability statements.

7.1 Where lemma exchange enters

The proof agents may discover normal forms, cancellation lemmas, or equivalent formulations of commutativity. A verified equivalence can simplify the finite-model constraints. Conversely, a partial model can reveal a small identity that separates abelian from nonabelian candidates and becomes a new target for P+ or P-. PI can package the successful separator into a general reduction. The model tables themselves remain semantic packets; only their verified consequences cross into other contexts.

7.2 What this example does not show

Finite model finding is not complete for arbitrary first-order theories, and set-theoretic independence usually cannot be witnessed by two small tables. The example validates the architecture, not its scale. A CH-level ALD would need formalized inner-model and forcing machinery; an arithmetic independence problem might require proof predicates, ordinal analysis, or consistency-strength arguments.

8. The shared search as a typed AND-OR hypergraph

Predator 8's proof search is naturally viewed as an AND-OR hypergraph. A goal node may have several alternative inference routes (OR), while a chosen inference may require several premises (AND). The ALD generalizes the node type from a formula to a context-formula pair.

Goal node: $G = (\text{context}, \text{formula}, \text{polarity}, \text{certificate type})$

Object edges remain ordinary formal inferences. Cross-context hyperedges are bridge theorems: weakening, deduction, proof coding, soundness, completeness, model satisfaction, interpretation, or forcing. A model pair is itself an AND edge: both model certificates and the soundness theorem are required before the INDEPENDENT node closes.

8.1 Local time and global time

With several agents, stage numbering must distinguish local expansions from global scheduler events. If $P+$ acts only on selected global stages, its state after global event m generally differs from the state of a single-target prover after m expansions. Comparisons must index each agent by its own local search steps, then separately account for scheduling and communication overhead. This repairs the temptation to equate alternating branches at identical global stage numbers.

8.2 Monotone stores, nonmonotone attention

The verified lemma stores can grow monotonically: once a theorem and certificate are accepted, they are not revoked absent a verifier or theory-version fault. A heuristic's attention is different. Adding a new candidate can change rankings and delay a previously preferred action. Logical availability is monotone; bounded heuristic performance need not be.

9. Generalizing cross-branch reuse

Two branches need one directional distinction: origin A versus origin B. Five agents require a matrix. Let the agent set be $I = \{P+, P-, PM+, PM-, PI\}$. For a verified packet λ , record every origin and every agent that makes an actual proof, model, or metatheoretic step using it.

Definition 9.1 (Use matrix)

For agents i and j , let $U_{ij}(B)$ be the number of packets originating at i and used by j within budget B . Let $O_i(B)$ be the number originating at i . Define $\kappa_{ij}(B) = U_{ij}(B) / O_i(B)$, with value 0 when $O_i(B) = 0$. Availability and actual use are logged separately, as in the original κ_{av}/κ_{us} distinction.

9.1 Fan-out and tri-use

Partition the agents into the positive class $\{P+\}$, negative class $\{P-\}$, and independence class $\{PM+, PM-, PI\}$. The fan-out $F(\lambda)$ is the number of distinct classes that actually consume λ . A **tri-use lemma** has $F(\lambda) = 3$. The tri-use rate $\kappa_{us}^{(3)}$ is the fraction of eligible verified packets consumed by all three classes. It measures the exact phenomenon motivating this sequel.

9.2 Settlement dividends

A reuse count does not by itself establish benefit. A crossed lemma may have an equally short substitute or may arrive too late. For each branch j , define its empirical dividend $\Delta_j(\lambda)$ by an ablation: rerun the frozen search with λ 's import disabled and measure the additional verified expansions, materializations, or wall time. The total dividend of λ is the sum of nonnegative branch dividends, reported together with any communication and verification cost.

$$\text{Utility}(\lambda) = [\sum_j w_j \Delta_j(\lambda)] / [\text{generation cost} + \text{verification cost} + \text{routing cost}]$$

9.3 A reuse matrix is diagnostic

A high $P \rightarrow P$ - cell and low model-to-proof cells would show that the system remains essentially a two-sided ATP. High model-to-PI reuse but zero proof-to-independence reuse would show that the semantic subsystem works while three-way exchange does not. A nonzero tri-use rate would be the strongest direct evidence that the ALD is behaving as a single collaborative logical system rather than five independent programs under one name.

10. Scheduling and the traffic-director problem

10.1 Fair dovetailing

The safest baseline is weighted round-robin scheduling. Every live branch receives a fixed positive share of expansions. In an unbounded conceptual run this is fair: no branch that requires finitely many local steps is starved forever. In a bounded experiment, the weights are frozen before the target is run and reported with the result.

10.2 Adaptive scheduling

A more capable coordinator may allocate extra effort to agents with high recent verified-lemma yield, high cross-class fan-out, shrinking model constraints, or a nearly complete certificate. Adaptive scheduling must retain an exploration floor. Otherwise early random success can monopolize the budget and suppress the very distant route that would have settled the conjecture.

10.3 Certified branch retirement

The coordinator may permanently stop a branch only for a logical reason or an operational stopping rule. A model of $T + \neg c$ can retire $P+$ because it certifies $T \not\models c$. A model of $T + c$ can retire $P-$. A complete proof or refutation settles the target immediately, unless the protocol continues other branches for diagnostic measurement. Low score, inactivity, or elapsed time may reduce a branch's allocation but may not be redescribed as a logical impossibility.

10.4 The coordinator is outside the trusted mathematical core

The scheduler can use machine learning, nondeterminism, language models, bandits, or hand-written rules. None must be trusted for correctness. The trusted surface consists of the certificate verifiers, theory identifiers, and destination-checked transports. This follows the certificate-verification architecture of Metamath [9] and the broader de Bruijn criterion: discovery may be complicated, while checking remains small and independent.

11. Creativity $Y = D N$ as a scheduling objective

The creativity article defines creativity as the product of distance D and novelty or productive value N . The product matters: a route that is conventional but useful has low D ; a route that is bizarre but useless has low N . Either near-zero factor drives Y toward zero. The ALD supplies an operational setting in which both factors can be measured rather than admired rhetorically.

11.1 A measurable distance score

$$D(\lambda) = \alpha d_{\text{representation}} + \beta d_{\text{theorem-graph}} + \gamma d_{\text{policy}} + \delta d_{\text{branch}}$$

Representation distance records a move such as logic to polynomial algebra, syntax to model theory, or object theory to metatheory. Theorem-graph distance records how far the supporting lemmas lie from the dominant route. Policy distance compares the generating tactic with the population's usual tactics. Branch distance rewards a packet that crosses from proof to model or metatheory rather than remaining inside one local search. Each component is normalized using the frozen training corpus, not tuned after seeing the benchmark result.

11.2 A measurable productive-novelty score

$$N(\lambda) = v(\lambda) [a \text{ fanout}(\lambda) + b \text{ verified-dividend}(\lambda) + c \text{ status-progress}(\lambda) + d \text{ robustness}(\lambda)]$$

The gate $v(\lambda)$ equals 1 only after verification and 0 otherwise. Fan-out measures how many branch classes use the packet. Verified dividend measures saved search under ablation. Status progress measures whether the packet completes or shortens a proof, model, consistency, or independence certificate. Robustness measures reuse across targets or seeds. A merely surprising statement with no verified consequence has N near zero.

11.3 How to maximize Y without destroying the search

The coordinator should not greedily select the largest observed Y . High Y is sparse, delayed, and noisy; greedy optimization can abandon ordinary steps needed to finish a proof. Use a portfolio. Reserve fixed capacity for reliable exploitation, fixed capacity for distant exploratory specialists, and adaptive capacity allocated by expected verified Y per unit cost. Diversity is maintained by multiple creativity profiles rather than one globally maximized parameter vector.

$$\text{score}(\text{agent } i) = p_{\text{solve}}(i) + \lambda E[Y_{\text{verified}}(i)] / E[\text{cost}(i)] + \text{fairness bonus}$$

11.4 The scientific question

Hypothesis 11.1 (Creativity-guided settlement)

Under a fixed anti-leakage protocol and equal resource budget, a population policy optimizing expected verified Y per cost solves some targets that a utility-only, novelty-only, or single-policy ATP does not.

This is a hypothesis, not a result. Maximizing Y could fail because D is measured badly, because useful proofs require long conventional prefixes, because model or metatheory libraries are missing, or because coordination overhead dominates. The right response is an ablation experiment, not a claim that formalized creativity must produce previously unreachable theorems.

12. A sound operational algorithm

The following pseudocode separates untrusted proposal from trusted acceptance. A packet may originate from any Predator, model finder, algebra system, or language model. It influences another agent only after its certificate and destination transport succeed.

```
INPUT: theory T, target c, metatheory M, budget B, frozen policies
initialize P+, P-, PM+, PM-, PI and typed stores
while budget remains:
    i := coordinator.select_agent_with_fairness()
    proposals := agent[i].advance_one_local_step()
    for packet in proposals:
        if source_verifier.rejects(packet): log REJECTED; continue
        store.insert(packet with context and provenance)
        for transport in registered_interfaces(packet):
            translated := transport.propose(packet)
            if destination_verifier.accepts(translated):
                destination_store.insert(translated)
                coordinator.notify_consumers(translated)
        status := assembler.check_complete_certificate_patterns()
        if status in {PROVED, REFUTED, INDEPENDENT, INCONSISTENT}:
            return status with complete certificate bundle and log
    return operational_stop_reason with no mathematical conclusion
```

12.1 Terminal certificate patterns

- A checked $\text{Obj}(T)$ proof of c closes PROVED.

- A checked $\text{Obj}(T)$ proof of $\neg c$ closes REFUTED.
- A checked $\text{Obj}(T)$ proof of contradiction closes INCONSISTENT.
- A checked $\text{Meta}(M)$ proof of both unprovability claims closes INDEPENDENT.
- Two checked opposite model or consistency packets plus the checked bridge theorem close INDEPENDENT.

12.2 Failure containment

A faulty agent can waste its own budget and flood the verifier with rejected packets. Rate limits and reputation scores control this operational problem. It cannot make an accepted theorem false unless it exploits a verifier, theory-identification, or certificate-translation defect. Those components therefore receive audit priority, version hashes, and independent regression tests.

13. Soundness, speed, and an important correction

Theorem 13.1 (ALD output soundness)

Assume: (i) the object and metatheory certificate verifiers are sound for their declared theories; (ii) every cross-context packet is accepted only after destination verification; and (iii) each model or consistency assembly rule used by the status assembler has a checked proof in M . Then every mathematical status returned by the ALD is correct relative to the declared theories and assumptions.

Proof. PROVED, REFUTED, and INCONSISTENT follow directly from the corresponding sound object verifier. For INDEPENDENT, the final packet is either a direct M -proof of both unprovability claims or an M -verified assembly from model, consistency, forcing, or interpretation certificates. By assumption the relevant bridge theorem is checked in M ; hence both unprovability claims hold relative to M . Destination verification and Theorem 4.2 prevent an invalid cross-context import from entering the assembly. $\square$

13.1 Sharing is sound more generally than it is fast

Once verified, an extra lemma cannot invalidate a proof system. It can nevertheless delay a bounded heuristic search by changing rankings, enlarging candidate queues, increasing verification load, or consuming memory. Therefore the statement 'sharing can never delay settlement' is valid only for an exhaustive extension-monotone expansion rule, a cost model that treats imports as free, or an architecture that preserves the baseline search separately. It is not unconditional for a selective heuristic.

Proposition 13.2 (Shadow-search non-delay guarantee)

Run an unchanged baseline branch beside the sharing-augmented branch and return the first verified certificate from either. Measured in baseline local expansions, sharing cannot delay the baseline proof. Measured in wall time or total resources, the guarantee includes the explicit shadow and communication overhead.

This shadow construction is the clean experimental control. A cheaper implementation may reserve a fixed fraction of each branch's budget for baseline actions, giving a bounded slowdown rather than strict non-delay. The original reuse theorem remains an excellent semantic observation - crossed closed lemmas are conservative derived rules - but a speed theorem needs a policy and cost hypothesis.

13.2 No false independence from dual failure

The assembler has no transition from $\{P+ \text{ failed}, P- \text{ failed}\}$ to INDEPENDENT. Only a metatheoretic certificate pattern can close that status. This syntactic design rule is stronger than a warning in the documentation: it makes the forbidden inference unrepresentable in the result state machine.

14. What no general ALD can do

Church and Turing proved that the general decision problem for sufficiently expressive logic has no algorithmic solution [6, 7]. Goedel's incompleteness theorems show that sufficiently strong, effectively axiomatized consistent theories leave sentences undecided and cannot internally certify all of the metatheoretic facts one might want [5]. These limitations do not make the ALD project incoherent. They determine its output vocabulary and its scope.

14.1 Partiality

A general ALD may run indefinitely in an ideal unbounded model or return an operational stop under a finite budget. It can recognize only independence results for which its chosen M , libraries, and search policies can produce a certificate. There will be true metatheoretic statuses beyond that reach.

14.2 Specialized total ALDs

Restricted domains can have complete decision procedures. Propositional logic, Presburger arithmetic, and various finite or decidable theories admit total classification algorithms under their standard formulations. In such a domain the ALD name can be literal. The architecture remains useful because proof and model certificates, portfolio search, and typed sharing can still improve explanation and practical performance.

14.3 Metatheoretic regress

When M certifies independence from T , one may ask what certifies M . The ALD does not eliminate foundations; it makes them explicit. A stronger metatheory M^+ can audit M , but no finite tower dissolves every foundational question. The practical goal is the same as in formal proof verification: minimize, expose, version, and independently check the trusted base.

15. Measurement protocol

The measurement protocol is not another logical branch. It is the experiment that decides whether the architecture actually helps. The original paper proposed measuring κ_{us} for two-way sharing. The ALD protocol generalizes that experiment to proof, refutation, model, and metatheory agents.

15.1 Freeze before running

- Exact versions and hashes of T , M , the verifier, all libraries, and all translations.
- The formal target c , its negation, parsing, definitions, and permitted background axioms.
- Training cutoff and anti-leakage exclusions for c , $\neg c$, known proofs, independence proofs, models, labels, and equivalent restatements.
- Hardware, memory, wall-clock and expansion budgets, overflow rules, thread count, and nondeterministic seed policy.
- Agent portfolio, scheduler weights, creativity profiles, and every criterion for branch retirement.

15.2 Controlled variants

VARIANT	ACTIVE COMPONENTS	QUESTION ANSWERED
A	P+ only	Single-target ATP baseline
B	P+ and P-, no sharing	Cost and benefit of symmetric settlement search
C	P+ and P- with typed Obj(T) sharing	Two-way cross-branch dividend
D	P+, P-, PI; no separate model specialists	Value of a minimal three-Predator ALD
E	Five Predators plus fixed fair coordinator	Value of semantic specialization
F	Variant E with $Y = D \cap N$ adaptive scheduling	Incremental effect of creativity guidance
G	Variant F with each sharing edge ablated in turn	Causal value of particular cross-context transports

15.3 Primary measurements

- Verified terminal status and certificate size; invalid certificates count as failure.
- Local expansions, materializations, verifier work, communication volume, wall time, and peak memory.
- Reuse matrices $\kappa_{ij,av}$ and $\kappa_{ij,us}$; fan-out distribution; tri-use rate $\kappa_{us}^{(3)}$.
- Branch and lemma dividends under ablation, plus coordination overhead.
- Solve rate across repeated seeds, not merely the best seed; confidence intervals and full outcome counts.
- Separate counts for PROVED, REFUTED, INDEPENDENT, INCONSISTENT, TIMEOUT, OVERFLOW, FAULT, and INTERRUPTED.

15.4 Publication discipline

Exploratory runs used to debug preprocessing, recover checkpoints, tune weights, or inspect the target must not enter the frozen publication table. An interrupted run may remain useful privately, but a clean benchmark requires a fresh preregistered run. For nondeterministic algorithms, publish the seed policy and outcome distribution. A claim that Predator proves something no other ATP can prove requires matched budgets, current comparison systems, repeated trials where applicable, and verified certificates from every claimed success.

Interpretation of the protocol

The experiment does not ask whether several Predators can talk. It asks whether their verified conversation changes settlement outcomes or resource use under a frozen budget.

16. Implementation roadmap for Predator

Predator 8.002 is not yet an ALD and should not be described as one. The shortest honest route is staged. Each phase should leave a testable artifact rather than waiting for the full independence system.

16.1 Phase 1: five object-level Predators and one shared pool

The first prototype deliberately ignores independence. Run five ATP workers inside the same frozen object theory T : two goal-directed workers for c , two for $\neg c$, and one lemma scout that searches for equivalences, normal forms, and structurally central intermediate theorems. Give the workers different search or creativity profiles. They all submit closed candidate lemmas to one verifier-gated Obj(T) pool and may consume every accepted pool entry.

- Five local search states sharing one parsed immutable corpus, but never sharing live substitutions.
- One verified Obj(T) lemma pool with origin, availability, actual use, dependencies, and proof hashes.
- A lightweight coordinator for scheduling, deduplication, routing, and measurement.

- Local expansion counters separated from global scheduler events and communication overhead.
- Single-worker, five-worker-no-sharing, and five-worker-sharing controls on nonduplicate targets.

This is not yet the full ALD because no agent can certify independence. It is the engineering and scientific foundation of the ALD: if five same-theory Predators cannot exchange lemmas safely and profitably, adding model theory and metatheory will only multiply the failure modes.

16.2 Phase 2: install typed contexts

- Theory and context IDs on every goal, lemma, proof, and cache entry.
- Destination verification for weakening, deduction, proof-code, and other transport adapters.
- Separate stores for $\text{Obj}(T)$, $\text{Obj}(T+c)$, $\text{Obj}(T+\neg c)$, $\text{Meta}(M)$, and model packets.

16.3 Phase 3: add finite-model specialists

Connect a model finder for a first-order fragment, but require a replayable model certificate: a finite domain, operation and relation tables, and a checker that evaluates every axiom and target sign. Prover9 and Mace4 historically illustrate complementary proof and finite-countermodel search [8], but an ALD must integrate the results into its own certificate discipline.

16.4 Phase 4: add the metatheoretic assembler

- Formalize the soundness bridge used by the selected fragment.
- Accept two opposite model certificates and construct a replayable independence bundle.
- Implement direct proof-predicate and consistency packets only after the relevant coding exists.

16.5 Phase 5: adaptive coordination and creativity

- Start with frozen fair weights; only then compare adaptive scheduling.
- Estimate D and N from training targets with the benchmark excluded.
- Run novelty-only, utility-only, product, and randomized controls.

16.6 Phase 6: set theory and genuine independence machinery

This is a separate research program. A set-theoretic ALD needs formal encodings of models of set theory, relative consistency, inner models, forcing or interpretations, and the associated preservation theorems. `set.mm` supplies a broad object-level foundation, but it does not automatically supply an arithmetized proof predicate for an arbitrary frozen subtheory or a ready-made forcing engine. The formalization burden is part of the theorem, not implementation plumbing.

17. Application to the halo benchmark and to new mathematics

17.1 The halo target

The benchmark $\text{Hal}(0) \approx R$ is intended as an object-level theorem in its frozen formal environment. $P+$ is therefore the branch expected to settle it. $P-$ can still generate useful cardinal, injection, order, or choice lemmas. PI can analyze axiom dependence: for example, whether a particular construction needs a choice principle, whether an alternative rational-function model avoids it, or whether a weakened base theory decides the formalized target. The independence machinery is useful even when the final status is **PROVED** because it maps where the theorem sits in a hierarchy of assumptions.

17.2 From known benchmarks to unknown theorems

A verified proof of a known hard target demonstrates search capability but not mathematical novelty. The next step is to freeze conjectures not present in training and not already settled in the literature, run the ALD, verify every output, and then conduct an independent novelty review. A machine-discovered proof settles the formal statement; a literature search and expert review establish whether humans had already settled it under another name.

17.3 Why the independence branch matters for discovery

Without PI, a difficult false target and a genuinely independent target can look identical to bounded proof search. With PI, the system can search for an explanation of nonsettlement rather than merely spend more budget. This widens the objective from 'prove theorems no human has proved' to 'settle conjectures no human has settled - by proof, refutation, or independence.' The third route does not weaken the first; its lemmas may be exactly what the first two routes need.

18. Open problems

The architecture is precise enough to fail in informative ways. The following questions are empirical or mathematical, not matters of terminology.

- Does $\kappa_{us}^{(3)}$ become nonzero on a real formal corpus under a real targeted policy?
- Which packet types most often cross from proof search into model or metatheory search?
- Can an equivalence or normal-form specialist systematically manufacture tri-use lemmas?
- Does a five-agent design outperform a three-agent design after verification and communication costs?
- Can shadow search or reserved baseline slots preserve performance while sharing remains aggressive?
- Does $Y = D \ N$ predict downstream verified dividends better than novelty or utility alone?
- For which fragments can model-pair independence certificates be reconstructed entirely in Metamath?
- What is the smallest metatheory sufficient to certify a given independence result?
- Can forcing and interpretation libraries be modularized into reusable proof-carrying interfaces?
- Do high tri-use rates correlate with unique solves, or merely with additional bookkeeping?

Conjecture 18.1 (Three-way settlement dividend)

There exists a frozen formal corpus, target family, and certificate-producing policy for which typed three-way lemma exchange among proof, refutation, and independence agents achieves a strictly greater verified solve rate under equal budget than every corresponding one-way or two-way ablation.

19. Conclusion

The Automated Logical Decider envisioned here is not a single omniscient prover. It is a disciplined society of proof-producing specialists. One Predator tries to prove c . A second tries to prove $\neg c$. Two semantic Predators try to construct opposite worlds in which c is true and false. A metatheoretic Predator tries to turn those worlds, consistency arguments, forcing constructions, or interpretations into a formal theorem of independence. A small coordinator directs traffic without being trusted as a mathematician.

The central new proposal is that these are not sealed searches. They exchange verified lemmas through typed interfaces. A theorem discovered while trying to prove c may simplify $\neg c$, constrain a model, and become the bridge that lets PI transfer an existing independence result. Conversely, a model can certify an unprovability direction and redirect proof-search resources. The three outcomes talk to one another without being logically confused.

The restraint is as important as the ambition. Theories and metatheories remain named. Certificates are checked in their destination contexts. Search failure never implies independence. Sharing is not assumed to improve bounded heuristic search; it is measured. Creativity $Y = D \ N$ is treated as a falsifiable scheduling

objective, not a guarantee. Under those rules, the ALD becomes a concrete research program: a machine that attempts not merely to prove conjectures, but to determine their formal status and to explain that status with a certificate.

The program in one sentence

Settle conjectures no human has settled - by proving them, refuting them, or proving their independence from explicitly named axioms - and let every route contribute verified lemmas to every other route.

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